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MATH6400: Quiz #8
Due: April 11, 2026 @ 23:59

The cubic equation $ax^3 + bx^2 + cx + d = 0$ can be converted to a *depressed cubic* $x^3 + px + q = 0$ by dividing by the leading coefficient a (which doesn't change the roots) and shifting so that the graph $y = x^3 + px + q$ has its inflection point on the y-axis. This shift is obtained by replacing x by $x - \frac{b}{3a}$.
The solution to the depressed cubic equation $x^3 + px + q = 0$ is

$$x = \sqrt[3]{-\frac{q}{2} + \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}}. \tag{1}$$

The above equation will be referred to as *the formula*.
Similar to how $b^2 - 4ac$ is called the *discriminant* when considering a quadratic equation $ax^2 + bx + c = 0$, we will call the expression

$$\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3$$

the *discriminant*. In class, we'll see that the following relationship holds:

$$\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3 \begin{cases} > 0 & \implies & 1 \text{ root} \\ = 0 & \implies & 2 \text{ roots}^* \\ < 0 & \implies & 3 \text{ roots.} \end{cases}$$

* There is a caveat when $p = q = 0$; in this case there is one solution.

1. Let $p(x) = x^3 - 6x - 40$, and use the formula (1) to find x such that $p(x) = 0$. Verify that the result you obtain from the formula <u>must</u> be 4.

   Solution: This cubic is already depressed *and sad*, so we can apply (1) directly. Observe the following

$$\begin{aligned} x &= \sqrt[3]{-\left(\frac{-40}{2}\right) + \sqrt{\left(\frac{-40}{2}\right)^2 + \left(\frac{-6}{3}\right)^3}} + \sqrt[3]{-\left(\frac{-40}{2}\right) - \sqrt{\left(\frac{-40}{2}\right)^2 + \left(\frac{-6}{3}\right)^3}} \\ &= \sqrt[3]{20 + \sqrt{400 - 8}} + \sqrt[3]{20 - \sqrt{400 - 8}} \\ &= \sqrt[3]{20 + \sqrt{392}} + \sqrt[3]{20 - \sqrt{392}} \\ &= \sqrt[3]{20 + 2\sqrt{98}} + \sqrt[3]{20 - 2\sqrt{98}} \\ &= \sqrt[3]{20 + 2\sqrt{2}\sqrt{49}} + \sqrt[3]{20 - 2\sqrt{2}\sqrt{49}} \\ &= \sqrt[3]{20 + 14\sqrt{2}} + \sqrt[3]{20 - 14\sqrt{2}} \end{aligned}$$

   At this point we need to exercise faith and assert the following:

$$\left(20 + 14\sqrt{2}\right)^{\frac{1}{3}} = a + \sqrt{2}b \tag{2}$$

$$\left(20 - 14\sqrt{2}\right)^{\frac{1}{3}} = a - \sqrt{2}b \tag{3}$$

   Focusing on (2), we can observe the following

$$\left(\left(20+14\sqrt{2}\right)^{\frac{1}{3}}\right)^3 = \left(a+\sqrt{2}b\right)^3 \Rightarrow 20+14\sqrt{2} = a^3+6ab^2+(3a^b+2b^3)\sqrt{2}$$

Let's guess $a=2, b=1$ and then check for equality

$$\begin{aligned}20+14\sqrt{2} &= (2)^3+6(2)(1)^2+(3(2)^2(1)+2(1)^3)\sqrt{2}\\ &= 8+12+14\sqrt{2}\\ &= 20+14\sqrt{2}\end{aligned}$$

From this we can see we have found an equivalent form. Now we solve $a+\sqrt{2}b+a-\sqrt{2}b$ where $a=2, b=1$ to see

$$2+\sqrt{2}+2-\sqrt{2} = 4\checkmark$$

2. Consider $x^3-6x-4=0$. Verify that the formula yields

$$\sqrt[3]{2+2\mathbf{i}}+\sqrt[3]{2-2\mathbf{i}},$$

and, by converting the numbers $2\pm 2\mathbf{i}$ into their *polar form* and exercising your trig muscles, that the three roots are $-2, 1+\sqrt{3}$, and $1-\sqrt{3}$.

Addendum

In class on 4/6, I attempted to illustrate by way of a computer-based example, how sad and lame the state of finding roots of polynomials is. It is not required for you to do so for this quiz, but I dare you to carry out the following prompts. At the very least, I hope you look at my responses using Maxima.

Use a computer (referred to henceforth as your *silicon colleague*) for this and some software that will do symbolic computations(doesn't matter which one; I use Maxima 'cuz it's free and I can right-click on code and get the LaTeX code to copy and paste).

1. Ask your silicon colleague to find the roots of $p(x) = x^5-2x^2+x-7$. Display the results as symbolic (if possible) and numeric (always possible).

2. Define $p(x)$ as a function to your silicon colleague and ask it to evaluate the function at each of the *numeric* roots it found. Display your results.